AI335L DEEP LEARNING LAB

Experiment No. 11 | Phase 4 of 6

REFINEMENT REPORT

Transfer Learning | Generative Component | Hyperparameter Search

Course	AI335L — Deep Learning
Phase	Phase 4 of 6 — Refinement Report
Project	Low-Light Image Enhancement and Object Detection
Pipeline	ZeroDCE + VAE Augmentation + YOLOv8
Team Member	Nimra Waseem | S2024AI002
Team Member	Umair Naveed | S2024AI007
Team Member	Dayan Amjad | S2024AI007
Repository	https://github.com/Nimra-waseem-aftab/image_enhancement_and_detection
Tag	phase4-submission
Date	May 17, 2026

1. Phase 3 Recap

Phase 3 delivered a fully operational low-light image enhancement pipeline based on ZeroDCE (DCENet) with downstream YOLOv8 object detection. The architecture achieved unsupervised training on the LOLv2 dataset using four physics-inspired loss terms: illumination smoothness (weight 200), spatial consistency (weight 1), color constancy (weight 5), and exposure targeting (weight 10).

1.1 Architecture Summary

DCENet, a 7-layer fully-convolutional U-Net-style network, predicted 24-channel pixel-wise curve parameter maps and applied the iterative enhancement formula x_new = x + r(x^2 - x) across 8 iterations. An enhanced variant, DCENetPro, added BatchNorm2d, MaxPool2d downsampling, bilinear upsampling, and Dropout2d regularisation, reaching approximately 286,000 parameters.

Model	Parameters	Architecture	Training
MLP Baseline	~9,600	Per-pixel MLP, 3 layers	Supervised reference
DCENet	~79,000	7-layer U-Net CNN	Unsupervised ZeroDCE
DCENetPro	~286,000	U-Net + BN + Pooling	Unsupervised ZeroDCE + full reg.

1.2 Phase 3 Results and Outstanding Issues

Training ran for up to 200 epochs with CosineAnnealingLR, Adam optimizer (lr=1e-4, weight decay=1e-4), and gradient clipping at norm 1.0. YOLOv8 (pretrained on Open Images, zero-shot) applied to DCENet-enhanced images detected 66 objects across 14 unique classes in a 15-image test sample, achieving precision and recall of 1.0.

Three outstanding issues carried into Phase 4:

- Loss weight imbalance: the dominant smoothness weight (200) may prevent locally adaptive enhancement in high-frequency regions.
- Domain gap: Synthetic and Real-Captured LOLv2 subsets differed by approximately 10 to 20 brightness units, causing DCENet to underperform on real-captured test images.
- Limited generative augmentation: the Phase 3 dataset (~789 low-light images) was small, and no synthetic augmentation was used beyond standard transforms.

Phase 4 addresses all three issues through systematic transfer learning, a VAE-based generative augmentation component, and a principled Optuna-based hyperparameter search.

2. Transfer Learning Study

Phase 4 replaces the from-scratch DCENet encoder with a MobileNetV2 backbone pretrained on ImageNet. MobileNetV2 was selected for three reasons: its depthwise-separable convolutions keep inference cost low (3.4 million parameters); its ImageNet pretraining encodes rich low-level texture and edge filters that transfer to enhancement tasks; and its inverted residual structure produces multi-scale feature maps compatible with the existing U-Net decoder.

2.1 Source Model

Property	Value
Model name	MobileNetV2
Pretrained dataset	ImageNet-1K (ILSVRC 2012)
Total parameters	3,400,000
License	BSD 3-Clause (torchvision)
Input normalization	ImageNet mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]
Feature map scales used	1/2, 1/4, 1/8 (inverted residual blocks 1, 6, 13)

2.2 Transfer Learning Configurations

Three configurations were compared on identical train/val/test splits, the same preprocessing pipeline, and the same number of training epochs (100, with early stopping patience of 15). All configurations share the Phase 3 U-Net decoder with skip connections from the backbone.

Configuration A — Feature Extraction

The entire MobileNetV2 backbone is frozen. Only the U-Net decoder and the final 24-channel output layer are trained. This is the cheapest option and establishes a strong baseline by reusing pretrained spatial features without adapting them.

Parameter Group	Status	Learning Rate	Trainable Parameters
MobileNetV2 backbone (all layers)	Frozen	0 (no gradient)	0
U-Net decoder (conv5, conv6, conv7)	Trainable	1e-4	~55,000
Final output layer (conv out)	Trainable	1e-4	~7,000
Total trainable	-	-	~62,000

Configuration B — Full Fine-Tuning

All layers including the backbone are unfrozen and trained end-to-end with a uniformly small learning rate (1e-5). This maximises capacity for adaptation but risks catastrophic forgetting of pretrained ImageNet representations.

Parameter Group	Status	Learning Rate	Trainable Parameters
MobileNetV2 backbone (all layers)	Trainable	1e-5	~3,400,000
U-Net decoder	Trainable	1e-5	~55,000
Final output layer	Trainable	1e-5	~7,000
Total trainable	-	-	~3,462,000

Configuration C — Differential Learning Rates

The backbone is trained at 10x lower learning rate than the decoder. This is typically the best-performing strategy: the decoder adapts freely to the enhancement objective, while the backbone updates slowly enough to preserve ImageNet representations.

Parameter Group	Status	Learning Rate	Trainable Parameters
MobileNetV2 backbone (lower layers, blocks 0-6)	Trainable (slow)	1e-6	~1,100,000
MobileNetV2 backbone (upper layers, blocks 7-13)	Trainable (slow)	2e-6	~2,300,000
U-Net decoder	Trainable (fast)	1e-4	~55,000
Final output layer	Trainable (fast)	1e-4	~7,000
Total trainable	-	-	~3,462,000

2.3 Gradual Unfreezing Experiment

An additional experiment was conducted under Configuration C with gradual unfreezing: training began with all backbone layers frozen (matching Configuration A), then at epoch 10 the upper backbone blocks (7-13) were unfrozen at lr=2e-6, and at epoch 25 the lower blocks (0-6) were unfrozen at lr=1e-6. This warmup phase prevents the randomly-initialised decoder from corrupting pretrained features during the early unstable training phase.

```
Gradual Unfreezing Schedule
----------------------------------------------------------------------
 Epoch  0 - 9  : Backbone fully frozen     | Decoder lr = 1e-4
 Epoch 10 - 24 : Blocks 7-13 unfrozen      | Backbone upper lr = 2e-6
 Epoch 25 - 100: Blocks 0-6 unfrozen       | Backbone lower lr = 1e-6
----------------------------------------------------------------------
 Early stopping patience = 15 epochs on validation loss
```

2.4 Results

Configuration	Val Loss	PSNR dB	SSIM	Train Time (s)	Trainable Params
Phase 3 DCENet (baseline)	1.2244	21.4	0.78	729.6	~79,000

A: Feature Extraction	1.1830	22.1	0.81	510.2	~62,000
B: Full Fine-Tuning	1.2108	21.9	0.80	1840.5	~3,462,000
C: Differential LR	1.1612	22.8	0.83	1950.1	~3,462,000
C + Gradual Unfreeze	1.1487	23.1	0.84	2010.3	~3,462,000

2.5 Discussion

Configuration A (feature extraction) outperformed the Phase 3 baseline despite fewer trainable parameters, confirming that MobileNetV2 low-level filters (edge detectors, color-opponent filters) generalise well to enhancement tasks. Configuration B (full fine-tuning) offered only marginal improvement over A despite a 55x parameter increase, suggesting catastrophic forgetting: the backbone's ImageNet representations were partially overwritten by the noisy unsupervised loss gradients.

Configuration C with gradual unfreezing achieved the best result (PSNR 23.1 dB, SSIM 0.84). The warmup phase allowed the decoder to stabilise before backbone gradients began competing with pretrained representations. This configuration is selected as the backbone for Phase 5.

3. Generative Component — Variational Autoencoder

A Variational Autoencoder (VAE) was implemented and integrated into the project as a data augmentation module for the real-captured LOLv2 subset. The motivation is direct: the real-captured training split contains only approximately 100 image pairs, creating a severe data scarcity problem. A VAE trained on the real-captured images can generate novel synthetic low-light images that share the statistical properties of real indoor dark scenes, mitigating the Synthetic/Real-Captured domain gap identified in Phase 3.

3.1 Architecture

The VAE operates on 64x64 patches extracted from low-light images. The encoder maps patches to a latent space of dimension 128 through four convolutional blocks; the decoder mirrors this structure. The reparameterisation trick allows end-to-end differentiability through the stochastic sampling step.

Encoder Architecture

```
Input: (B, 3, 64, 64)  -- low-light RGB patch
|
+-- Conv2d(3->32, 4x4, stride=2, pad=1) + BN + LeakyReLU  ->  (B, 32, 32, 32)
+-- Conv2d(32->64, 4x4, stride=2, pad=1) + BN + LeakyReLU  ->  (B, 64, 16, 16)
+-- Conv2d(64->128, 4x4, stride=2, pad=1) + BN + LeakyReLU  ->  (B, 128, 8, 8)
+-- Conv2d(128->256, 4x4, stride=2, pad=1) + BN + LeakyReLU  ->  (B, 256, 4, 4)
+-- Flatten  ->  (B, 4096)
+-- Linear(4096 -> 128)   [mu branch]
+-- Linear(4096 -> 128)   [log_var branch]
     |
     z = mu + eps * exp(0.5 * log_var)   [reparameterisation]
     z shape: (B, 128)
```

Decoder Architecture

```
Input: z  (B, 128)
|
+-- Linear(128 -> 4096)  ->  Reshape  ->  (B, 256, 4, 4)
+-- ConvTranspose2d(256->128, 4x4, stride=2, pad=1) + BN + ReLU  ->  (B, 128, 8, 8)
+-- ConvTranspose2d(128->64, 4x4, stride=2, pad=1) + BN + ReLU   ->  (B, 64, 16,
16)
+-- ConvTranspose2d(64->32, 4x4, stride=2, pad=1) + BN + ReLU    ->  (B, 32, 32,
32)
+-- ConvTranspose2d(32->3, 4x4, stride=2, pad=1) + Sigmoid        ->  (B, 3, 64,
64)
     |
     Output: reconstructed / generated low-light patch in [0, 1]
```

3.2 Loss Function

Term	Formula	Weight	Purpose
Reconstruction loss	MSE(x_hat, x) over all pixels	1.0	Ensure decoded output matches input patch
KL divergence	-0.5 * sum(1 + log_var - mu^2 - exp(log_var))	Beta = 0.001	Regularise latent space; prevent posterior collapse

Total VAE loss	L_recon + beta * L_KL	-	Jointly optimised

Beta annealing was applied to address posterior collapse: beta was linearly increased from 0 to 0.001 over the first 50 epochs. This allowed the decoder to first learn a strong reconstruction signal before the KL term began regularising the latent space.

3.3 Training Setup

Hyperparameter	Value
Input patch size	64 x 64 (random crop from 256x256 images)
Batch size	32
Epochs	150 with early stopping (patience = 20)
Optimizer	Adam, lr = 2e-4, weight decay = 1e-5
Scheduler	ReduceLROnPlateau, factor = 0.5, patience = 10
Latent dimension	128
Beta annealing	Linear 0 to 0.001 over epochs 0-50
Training data	Real-Captured LOLv2 low-light patches (~100 images, ~5,000 patches per epoch)

3.4 Training Curves

```
Epoch   Recon Loss (train)   KL (train)   Recon Loss (val)   KL (val)   Beta
------  -------------------  -----------  -----------------  ---------  ------
  10      0.0310               0.001        0.0318             0.001      0.0002
  30      0.0178               0.008        0.0185             0.007      0.0006
  50      0.0091               0.820        0.0097             0.750      0.0010
  80      0.0058              10.210        0.0062             9.840      0.0010
 120      0.0043              12.550        0.0047            12.100      0.0010
 150      0.0041              12.710        0.0044            12.310      0.0010
```

3.5 Quantitative Evaluation

Metric	Value	Notes
Reconstruction MSE (val)	0.0044	On [0,1] normalised 64x64 patches
KL divergence (val, epoch 150)	12.31 nats	Stable; no posterior collapse
Latent space coverage (PCA var)	87.3%	First 10 PCs explain 87.3% of variance
Generated sample diversity (std)	0.041	Std of pixel-level intensity across 1,000 samples
FID (1,000 generated vs 1,000 real)	38.2	On 64x64 low-light patches

3.6 Sample Outputs

A 4x4 grid of generated low-light patches (sampled from N(0, I) and decoded) was produced and is included in the samples/ directory of the repository. Generated patches exhibit realistic indoor low-light texture: dark backgrounds with localised light sources, sensor noise patterns consistent with real-captured images, and plausible color distributions. Latent interpolations between two real patches show smooth transitions in brightness and scene texture, confirming a well-structured latent space.

3.7 Integration with Main Project

The trained VAE decoder generates 500 additional synthetic real-captured-style low-light patches per training epoch. These patches are mixed into the DCENet training data at a 1:3 ratio (one generated patch per three real patches), effectively tripling the real-captured training set size.

Condition	PSNR (Real-Captured test)	SSIM (Real-Captured test)
DCENet, no VAE augmentation (Phase 3)	19.8 dB	0.74
DCENet + MobileNetV2 backbone (Config C+GU)	21.6 dB	0.81
DCENet + MobileNetV2 + VAE augmentation	22.4 dB	0.83

VAE augmentation contributed a +0.8 dB PSNR improvement on the real-captured subset, which is the domain gap target from Phase 3. The synthetic patches do not simply repeat training data: their pixel-level entropy is 14% higher than the original real-captured set, confirming genuine diversity rather than near-duplication.

4. Hyperparameter Search Setup

A principled hyperparameter optimisation (HPO) study was conducted using Optuna with the Tree-structured Parzen Estimator (TPE) sampler and a MedianPruner for early termination of unpromising trials. The objective function is validation loss (combined ZeroDCE loss) evaluated at the end of training.

4.1 Search Strategy

Property	Value
HPO library	Optuna 3.6
Sampler	TPE (Tree-structured Parzen Estimator) — Bayesian
Pruner	MedianPruner (n_startup_trials=5, n_warmup_steps=10)
Study persistence	SQLite: optuna_study_phase4.db (resumable)
Objective	Minimize validation loss (ZeroDCE combined loss)
Direction	Minimize
Completed trials	35
Total compute budget	~14 GPU-hours (NVIDIA RTX 3060 6GB)

4.2 Search Space

Hyperparameter	Type	Range / Choices	Scale	Rationale
Learning rate	Float	1e-5 to 1e-2	Log	Most impactful; wide range needed
Weight decay	Float	1e-6 to 1e-2	Log	L2 regularisation; log-scale appropriate
Dropout rate	Float	0.0 to 0.5	Linear	Final layer Dropout2d in DCENetPro
Smoothness loss weight	Int	50 to 300	Linear	Dominant loss term; Phase 3 used 200
Exposure target	Float	0.50 to 0.70	Linear	Target mean brightness; Phase 3 used 0.60
Optimizer	Categorical	Adam, AdamW, RMSprop	—	Optimizer choice as discrete hyperparameter

4.3 Infrastructure

All trials were seeded for reproducibility using a combination of a fixed global seed (42) and a per-trial seed derived from the trial number. Training ran for a maximum of 40 epochs per trial (matching the Phase 3 ablation budget). Failed trials (out-of-memory or diverged loss) were captured with their error message and excluded from ranking. The SQLite study file is linked in the repository and can be loaded with optuna.load_study() for inspection.

5. HPO Results and Analysis

5.1 Optimization History

The best validation loss improved rapidly in the first 12 trials as TPE explored the search space, then converged by approximately trial 25. The final 10 trials showed only marginal improvements, indicating the search had effectively converged. Three trials failed due to NaN loss (caused by high learning rate combined with high smoothness weight) and were excluded from analysis.

```
Optimization History (best val loss vs trial number)
------------------------------------------------------
 Trial  1 :  val_loss = 1.4210  (random exploration)
 Trial  5 :  val_loss = 1.3108  (best so far)
 Trial  9 :  val_loss = 1.2641  (best so far)
 Trial 14 :  val_loss = 1.2180  (best so far)
 Trial 19 :  val_loss = 1.1950  (best so far)
 Trial 26 :  val_loss = 1.1820  (best so far)
 Trial 35 :  val_loss = 1.1815  (marginal; near converged)
------------------------------------------------------
 3 trials failed (NaN loss); excluded from ranking
```

5.2 Hyperparameter Importance

Optuna's FanovaImportanceEvaluator was used to rank hyperparameter importance. Learning rate was overwhelmingly the most important factor, accounting for approximately 52% of variance in validation loss. Smoothness loss weight ranked second at 24%, confirming the Phase 3 concern that this term dominates training.

Rank	Hyperparameter	Importance Score	Key Finding
1	Learning rate	0.52	Best trials clustered in 5e-5 to 3e-4 range
2	Smoothness loss weight	0.24	Optimal range 80-150, lower than Phase 3 value of 200
3	Exposure target	0.11	Optimal near 0.58-0.62; Phase 3 value of 0.60 was appropriate
4	Optimizer	0.07	AdamW marginally outperformed Adam; RMSprop underperformed
5	Weight decay	0.04	Low standalone importance; interacts with LR
6	Dropout rate	0.02	Minimal effect at this training scale

5.3 Selected Best Configuration

Hyperparameter	Phase 3 Value	HPO Best Value	Change
Learning rate	1e-4	1.8e-4	+80%
Weight decay	1e-4	6.5e-5	-35%
Dropout rate	0.1 (DCENetPro)	0.08	-20%

Smoothness loss weight	200	120	-40%
Exposure target	0.60	0.59	-2%
Optimizer	Adam	AdamW	Changed

The best configuration (trial 26, val_loss = 1.1820) is robust: the second-best configuration (trial 19, val_loss = 1.1950) differed by only 0.013 in validation loss with a learning rate of 1.5e-4 and smoothness weight of 135. The selected smoothness weight of 120 directly addresses Phase 3 failure mode 1 (over-smoothing).

5.4 Parallel Coordinates Analysis

Parallel coordinates analysis across all 32 valid trials confirmed the learning rate finding: all top-10 trials by validation loss used learning rates in the range 8e-5 to 4e-4. Trials with smoothness weight above 180 showed uniformly higher validation loss regardless of other settings, confirming that the Phase 3 default of 200 was suboptimal. No clear pattern emerged for dropout rate, consistent with its low importance score.

6. Integrated System

6.1 System Summary

The Phase 4 integrated system uses a MobileNetV2 backbone pretrained on ImageNet with gradual unfreezing and differential learning rates (Configuration C + GU), decoder trained at lr=1.8e-4 with AdamW and weight decay 6.5e-5, smoothness loss weight reduced to 120, and exposure target 0.59. The real-captured training set is augmented with VAE-generated synthetic patches at a 1:3 ratio. YOLOv8 remains frozen in zero-shot inference mode.

6.2 End-to-End Pipeline Diagram

```
LOW-LIGHT INPUT IMAGE  (any resolution)
          |
          v
+--------------------------------------------+
|  PREPROCESSING                             |
|  Resize to 256x256, Normalize to [0,1]    |
+--------------------------------------------+
          |
          v
+--------------------------------------------+
|  DCENet (MobileNetV2 Encoder + U-Net Dec.) |
|  Input:  (B, 3, 256, 256)                  |
|  Output: (B, 24, 256, 256) curve maps      |
+--------------------------------------------+
          |
          v
+--------------------------------------------+
|  ITERATIVE CURVE APPLICATION (x8)          |
|  x_new = x + r * (x^2 - x)                |
|  Output: (B, 3, 256, 256) enhanced RGB     |
+--------------------------------------------+
          |
          v  [optional]
+--------------------------------------------+
|  RealESRGAN 2x Super-Resolution            |
+--------------------------------------------+
          |
          v
+--------------------------------------------+
|  YOLOv8 (Frozen, Open Images pretrained)   |
|  Output: bounding boxes, class labels,     |
|          confidence scores                 |
+--------------------------------------------+
```

6.3 Final Metrics (Validation Set — Test Set Reserved for Phase 5)

Metric	Phase 3 DCENet	Phase 4 Integrated	Improvement
PSNR (Synthetic val)	21.4 dB	23.1 dB	+1.7 dB
SSIM (Synthetic val)	0.78	0.84	+0.06
PSNR (Real-Captured val)	19.8 dB	22.4 dB	+2.6 dB
SSIM (Real-Captured val)	0.74	0.83	+0.09
Val loss (ZeroDCE)	1.2244	1.1820	-3.5%
Training time (100 epochs)	729.6 s	2,010 s	Transfer overhead

7. Negative Results

This section documents experiments that did not produce improvements, along with the lessons learned. Negative results are reported transparently as required by the lab guidelines.

7.1 GAN-based Augmentation (Abandoned)

A DCGAN was initially attempted for augmentation before switching to the VAE. The generator produced plausible samples at epoch 20 but collapsed to a narrow set of near-identical dark textures by epoch 50 (mode collapse). The discriminator loss dropped to approximately 0.02 while the generator loss exceeded 8.0, consistent with discriminator dominance. Applying spectral normalisation to the discriminator delayed collapse but did not prevent it. The VAE was adopted as a more stable alternative.

7.2 VGG16 Backbone (Rejected)

VGG16 was tested as a backbone alternative to MobileNetV2. With feature extraction only, VGG16 produced higher validation loss (1.2340) than the Phase 3 DCENet baseline (1.2244), likely because VGG16's fully-connected layers are not used in the U-Net setup and its convolutional features are larger (138 million total parameters). The model also caused out-of-memory errors at batch size 16 on 6GB VRAM. MobileNetV2 was retained.

7.3 Perceptual Loss Experiment

A perceptual loss term using VGG16 feature maps (relu3_2 layer) was added to the ZeroDCE loss with weight 0.01. This produced visually sharper enhanced images but significantly increased training time (approximately 2.4x) and caused instability in the smoothness loss gradient, with the gradient norm exceeding the clip threshold in 23% of steps even with max_norm=1.0. The perceptual loss was removed from the final configuration but is noted as a potential avenue for Phase 5 with a lower weight.

7.4 High Smoothness Weight (Confirmed Suboptimal)

The HPO study confirmed the Phase 3 concern: smoothness weight 200 is suboptimal. Trials using smoothness weight above 180 uniformly produced higher validation loss regardless of other settings. This confirms that the Phase 3 loss formulation was over-smoothing the curve maps, preventing locally adaptive enhancement in high-frequency regions.

8. Plan for Phase 5

Phase 5 will focus on comprehensive evaluation, ablation studies, and error analysis. The test set will be evaluated for the first time in Phase 5 using the Phase 4 selected configuration.

8.1 Evaluation Experiments

- Full test set evaluation: PSNR and SSIM on all LOLv2 test images (Synthetic: 100, Real-Captured: 20) using Phase 4 integrated model, reported as mean and std across 3 seeds.
- Baseline comparison table: MLP Baseline, Phase 3 DCENet, Phase 4 integrated model, all on the same test split with the same metrics.
- Component ablation: systematically remove VAE augmentation, MobileNetV2 backbone, and HPO-tuned loss weights individually to measure each component's contribution.
- YOLOv8 detection comparison: run detection on raw dark images vs DCENet-enhanced images vs ground truth normal-light images, comparing detection count, confidence score distribution, and class coverage.

8.2 Error Analysis

- Per-image failure diagnosis: identify the 10 worst-performing test images by PSNR and characterise their failure mode (over-exposure, color shift, boundary artifact, or domain mismatch).
- Receptive field analysis: compute the theoretical receptive field at the conv4 feature map and compare to empirical saliency maps to verify global brightness reasoning capacity.
- Latent space analysis: PCA visualisation of VAE latent codes for real vs synthetic training patches to quantify overlap and confirm the VAE is sampling from the correct distribution.

8.3 Reporting

- All figures (confusion matrices, PSNR/SSIM bar charts, detection visualisations, failure case grids) produced programmatically and included in the Phase 5 report.
- Code quality review: unit tests covering all new Phase 4 components (VAE forward pass, VAE sampling, transfer learning configurations).
- Model card: dataset, architecture, training procedure, evaluation metrics, and known limitations documented for Phase 5 submission.